Chapter 4 Smart Manufacturing, Robotics, and AI Systems

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4.1 Introduction to Smart Manufacturing

Smart manufacturing signifies the future of the manufacturing industry, enhancing production intelligence through the integration of advanced information technologies and artificial intelligence (AI). This section begins with an overview of how manufacturing paradigms have evolved – from individual craftsmanship to Industry 4.0 – underscoring the pivotal role of digitalization and connectivity in modern manufacturing. It then delves into the key components of smart manufacturing, including the Internet of Things (IoT), big data analytics, cloud computing, and cyber-physical systems (CPS), which collectively drive efficiency improvements and cost reductions. The convergence and application of these technologies not only optimize production processes but also lay the groundwork for continuous innovation and sustainable development in manufacturing.

4.1.1 Evolution of Manufacturing Paradigms

Historically, the trajectory of manufacturing began with individual craftsmanship and evolved through several stages marked by revolutionary technological advancements (Schwab 2017). The first industrial revolution ushered in mechanical production facilities, followed by the second revolution's focus on mass production powered by electrical energy. The third industrial revolution brought automation through electronics and IT systems, paving the way for mass customization. Presently, we are amid the fourth wave, Industry 4.0, where digitization, AI, and interconnectivity drive manufacturing innovation. As a key offshoot of Industry 4.0, smart manufacturing places a strong emphasis on personalized production, notably through additive manufacturing techniques (Yao and Lin 2016).

4.1.2 Industry 4.0

Industry 4.0 marks the dawn of the Fourth Industrial Revolution, setting itself apart through a profound emphasis on interconnectivity, automation, ML, and real-time data analytics (Lasi et al. 2014). This era ushers in a significant shift toward the development of autonomous systems

and the facilitation of seamless data exchange within the realm of manufacturing technologies. As a result, this transformative phase paves the way for the evolution of "smart factories." These modern manufacturing units are adept at utilizing CPS to meticulously monitor and analyze physical processes. By doing so, they not only craft a dynamic, virtual replica of the tangible world but also excel in making decentralized decisions (Xu et al. 2018). This innovative approach enhances efficiency, flexibility, and quality in manufacturing, signifying a monumental leap toward an interconnected, intelligent production ecosystem. Viewed through this lens, Industry 4.0 heralds a new age of industrialization, where the blend of advanced technological frameworks promises to redefine the landscapes of production and operations across various sectors.

4.1.3 Key Components of Smart Manufacturing/Industry 4.0

Smart manufacturing, commonly referred to as Industry 4.0, ushers in a new era in industrial evolution (Yao et al. 2019b), brimming with the intelligence of next-generation technologies (Yao et al. 2017). It represents a transformative surge powered by advancements such as the IoT, cloud computing, CPS, advanced robotics, and ML with a notable emphasis on deep learning. Alongside are big data analytics (Yao et al. 2019b) and three-dimensional (3D) printing technologies (Yao and Lin 2016). The essence of realizing Industry 4.0 lies in the seamless integration of foundational technological principles, encompassing IoT, CPS, additive manufacturing, vast datasets, cybersecurity measures, cloud infrastructure, and more (Cañas et al. 2021). The four principal technological cornerstones, which are detailed subsequently, form the bedrock upon which Industry 4.0 is built.

1. IoT: Central to smart manufacturing, IoT a networks embed sensors and actuators within physical objects to facilitate data collection and exchange (Atzori et al. 2010). This connectivity enables enhanced monitoring, control, and optimization of manufacturing operations and processes, creating substantial value for businesses.
2. Big Data Analytics: With an immense volume of data generated by IoT, big data analytics becomes an essential tool for extracting actionable insights (Manyika 2011). The application of such analytics in manufac-

turing processes enhances quality control, supply chain efficiency, prediction of machine failures, and reduction of production downtime (Lee et al. 2013).

3. Cloud Computing and Internet of Services (IoS): Cloud computing has revolutionized the way data and services are managed, providing scalable resources and Software as a Service (SaaS) to manufacturers (Mell and Grance 2011). Coupled with IoS, it empowers businesses to offer new and innovative services as well as advanced capabilities in manufacturing (Wu et al. 2013).
4. CPS: It integrates computational algorithms and physical components, extending the capabilities of human operators and allowing for new forms of human-to-machine and machine-to-machine interactions (Lee 2008). Within manufacturing, they enable functions such as real-time monitoring, analysis, and diagnostics at an unparalleled level.

4.2 Smart Manufacturing System Integration and Interoperability

The concept of systems integration and interoperability forms the cornerstone of smart manufacturing. The seamless exchange of information and the harmonious working of diverse systems are pivotal in realizing the full potential of Industry 4.0. This section delves into the systematic orchestration of smart manufacturing components by introducing reference models, exploring the intricacies of cyber-physical production systems (CPPS), examining their extended versions, and addressing the role of robotics and AI at this intersection.

For smart manufacturing systems to actualize their full potential, integration and interoperability among their constitutive technologies – IoT, internet by and for people (IbfP)/internet of people (IoP), and internet of contents and knowledge (IoCK), big data analytics, cloud computing, CPS, robotics, and AI – are imperative (Yao et al. 2020). The frameworks and concepts explored here provide a blueprint for such harmonization, which is crucial for the advancement of industrial processes and productivity.

4.2.1 Smart Manufacturing Reference Model

To achieve integration and interoperability in smart manufacturing, a reference model is essential. Such a framework outlines the standardized protocols, interfaces, and architectures needed for the various elements of smart manufacturing systems to communicate and operate collaboratively. The Reference Architectural Model Industry 4.0 (RAMI 4.0) (Hankel 2015) is one such model that provides a structured approach to integration, covering aspects from business processes to the physical components. RAMI 4.0 organizes these aspects into a 3D map that includes layers for hardware and software, the product lifecycle value stream, and hierarchy levels defined by the International Electrotechnical Commission (IEC) (Nakagawa et al. 2021).

Industry 4.0 and smart manufacturing strive to achieve three types of integrations (Alcácer and Cruz-Machado 2019): 1) Vertical integration fosters connectivity and communication across various hierarchical levels within a single enterprise, optimizing processes and data flow; 2) Horizontal integration emphasizes collaboration and cohesion among separate entities, nurturing interconnected ecosystems that enhance cooperation and productivity; 3) End-to-end integration transcends the boundaries of production, encompassing the entire value chain from the initial customer interaction to the final delivery of products or services delivery.

4.2.2 Extended Smart Manufacturing System Integration and Interoperability

Wisdom manufacturing represents an advancement of socio-CPPS (SCPPS) that expands beyond the realm of smart manufacturing into the CPPS domain (Yao et al. 2019b). By integrating IoT, IoS, IbfP/IoP, and IoCK (Ma et al. 2022b). Similar to RAMI 4.0, the Reference Architectural Model for Wisdom Manufacturing (RAM4WM), illustrated in Figure 4.1 (Yao et al. 2024), organizing interoperability layers into six abstract layers: asset, communication, data, function, business, and users, arranged from bottom to top. These layers correspond to six organizational semiotic levels: physical, empirical, syntactic, semantic, pragmatic, and social, respective-

ly, providing a comprehensive framework for both understanding and implementation.

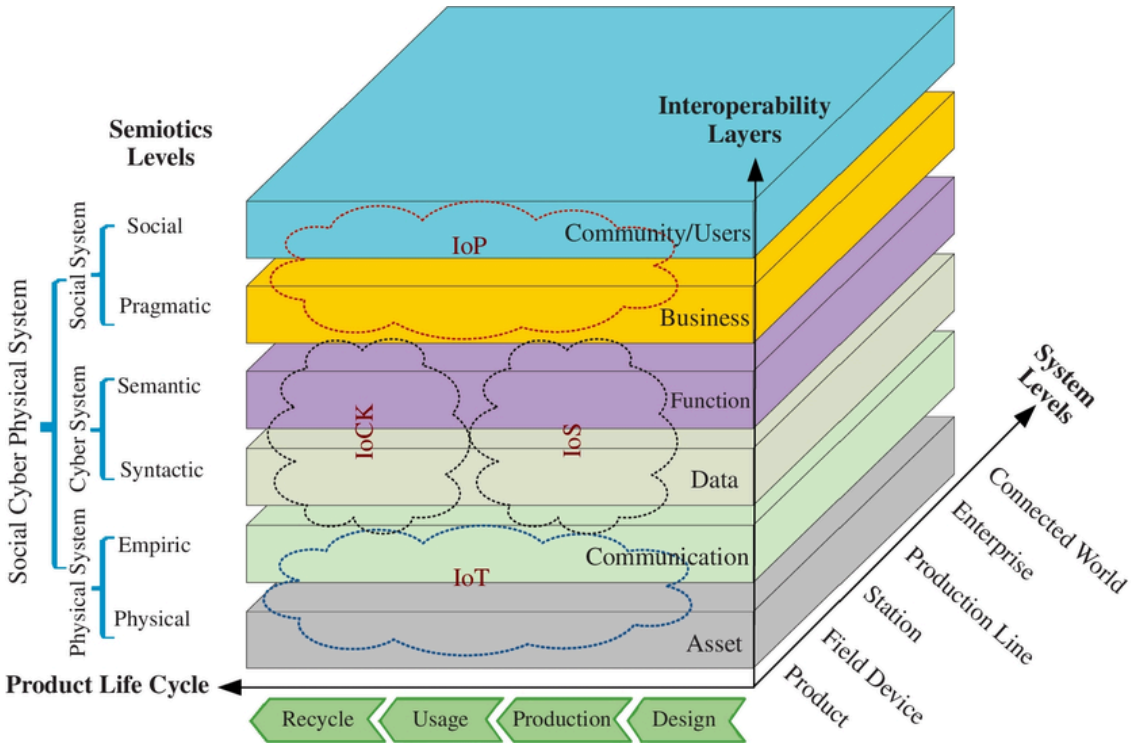


Figure 4.1 Reference Architectural Model for Wisdom Manufacturing (RAM4WM).

Source: Xifan Yao.

As an evolution beyond smart manufacturing, wisdom manufacturing adopts a loosely coupled service-oriented architecture, diverging from the conventional tightly coupled hierarchical manufacturing architecture (Yao et al. 2022b). This approach serves to reconcile the divide between information technology (IT) and operational technology (OT) within and/or among enterprises, thereby fostering integration and synergy across traditionally distinct domains. Such a transition approach not only redefines technological infrastructure but also reshapes collaboration among various operational facet, promoting a more interconnected and responsive framework for modern manufacturing.

4.2.3 Cyber-physical Production Systems

At the core of smart manufacturing resides the concept of CPPS (Yao et al. 2019b). These sophisticated systems intertwine computational algorithms with physical components, empowering real-time analysis, diagnosis, and task execution, thus amplifying efficiency and adaptability within manu-

facturing processes (Monostori 2014). CPPS transcends conventional automation, demanding the seamless integration of components like sensors, actuators, control systems, and software. This integration serves as a foundation for an interconnected and smart manufacturing environment that optimizes production capabilities and responsiveness.

4.2.4 Extended CPPS

The realm of CPPS has expanded to envelop more sophisticated and ambitious manufacturing systems and concepts. This extended CPPS (XCPPS) framework envisions a manufacturing ecosystem that is inclusive of advanced human collaboration, socio-technical systems, and even virtual environments.

4.2.4.1 Socio-cyber-physical Production System

SCPPS integrates the social aspects of manufacturing systems by examining the interaction between humans and machines, and the impact on society (Yao et al. 2014). It emphasizes the importance of social intelligence in smart manufacturing (Yao et al. 2019b), promoting machines to adapt with human requirements and practices.

4.2.4.2 Autonomous Cyber-physical Production Systems

This subsection delves into the concept of autonomous cyber-physical production systems (ACPPS) wherein systems are engineered to function seamlessly without human intervention. Such systems leverage AI and ML to autonomously monitor production processes, adjusting to variations, refining operations, and self-diagnosing problems. Nevertheless, even within an entirely ACPPS, there exists the framework of human-on-the-loop (HoL) and human-out-of-the-loop (HooL), if not human-in-the-loop (HiL) (Yao et al. 2022a) to maintain oversight and ensure safety and reliability.

4.2.4.3 Human-centric CPPS

Whereas SCPPS underlined social integration, human-centric CPPS (HCPPS) targets the optimization of human–machine interaction for im-

proved productivity and ergonomic working conditions. This perspective centralizes human roles and acknowledges operators' value and expertise within increasingly automated production systems, thereby driving innovation in human-aided ML and assistive technologies that pave the way for Industry 5.0 (Ma et al. 2022a).

4.2.4.4 Metaverse CPPS

Among the most speculative and futuristic of XCPPS variants is the Metaverse CPPS (Yao et al. 2024). Envisioned as an SCPPS operating within a virtual universe, these environments are the prototypes for testing, simulating, and managing manufacturing processes within entirely virtual domains before applying them in the real world – a concept that extends the idea of digital twins to new heights.

At the heart of these XCPPS variants lies SCPPS, embodying a sociotechnical system referred to as SCPPS. This core concept drives the evolution of CPPS toward a human-centric, environmentally friendly, and resilient manufacturing approach tailored for Industry 5.0.

4.2.5 The Intersection of Robotics and AI in Manufacturing

The synergy of robotics and AI plays a pivotal role in elevating the interoperability and integration within smart manufacturing systems. Through the infusion of AI, robots gain the capacity to execute intricate tasks, assimilate knowledge from their surroundings, and enhance their operations iteratively. Collaborative robots, often referred to as cobots, operate alongside human operators, assimilating insights from human actions, and providing invaluable support that amplifies not only efficiency but also safety within manufacturing environments. Moreover, AI algorithms meticulously scrutinize data generated by robots and other interconnected devices to optimize the entire manufacturing process, prognosticate maintenance needs, and dynamically adapt to new conditions without necessitating manual intervention (Figure 4.2), as detailed further below.

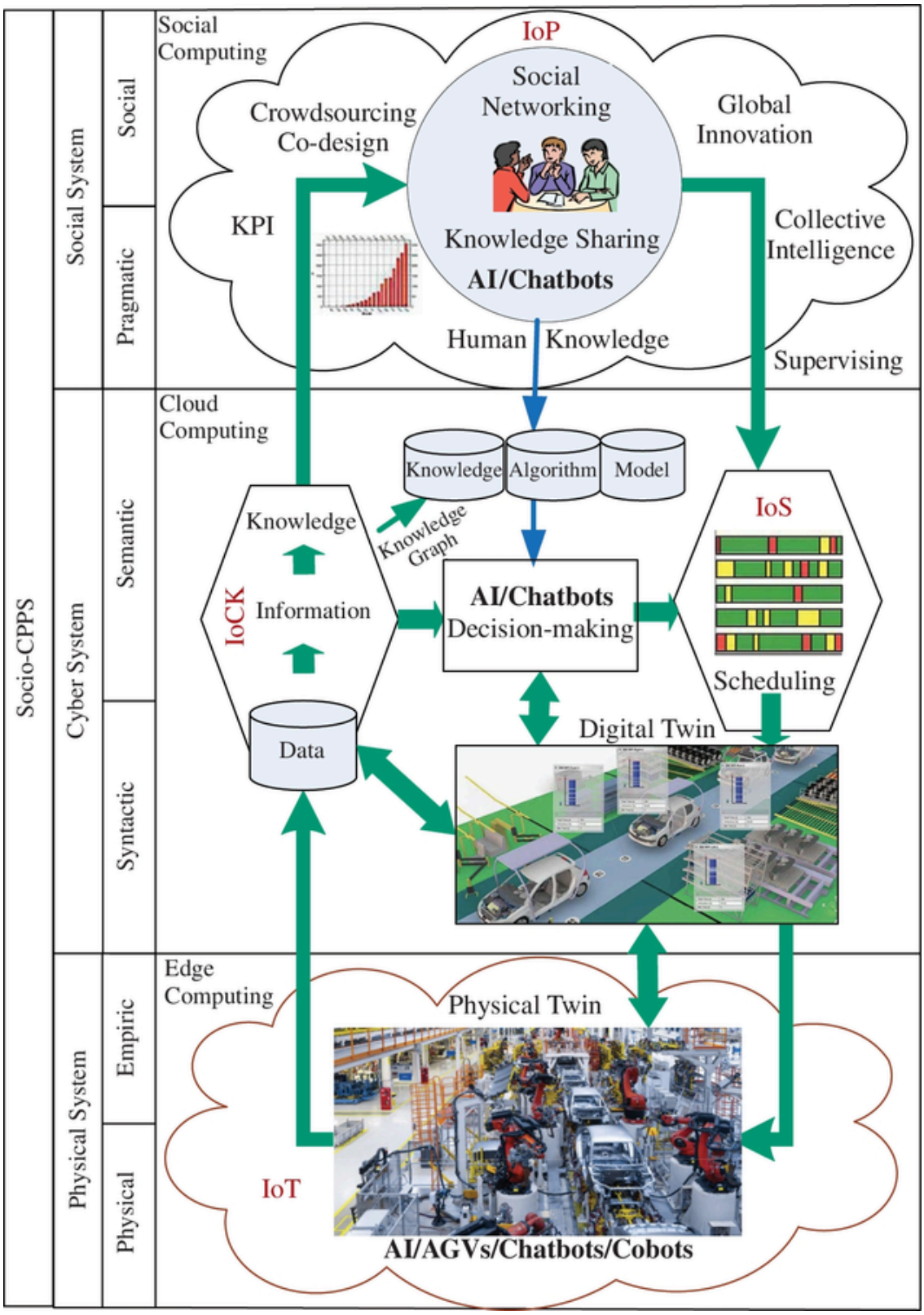


Figure 4.2 Robotics and AI synergized in SCPPS.

Source: Xifan Yao; Ivan Traimak/Adobe Stock Photos.

4.3 Robotics in Manufacturing

As the Fourth Industrial Revolution gathers momentum, robotics stands as a pivotal force in the transformation of the manufacturing industry. This section delves into the role of robots in modern manufacturing, trac-

ing their evolution from historical perspectives to the diverse types of robots in use today, and examining their integration with AI to enhance production efficiency and safety.

4.3.1 The Rise of Robotics in Manufacturing

The history of robots in manufacturing traces back to the 1960s with the introduction of UNIMATE, the first industrial robot to hit production lines, a milestone leap toward automation (Mickle 1961). Technological advancements since then have translated robotics from theory to practicality in manufacturing.

Between the 1970s and 1990s, the integration of sensor technology empowered robots to engage more intricately with their surroundings. Early breakthroughs like Shakey (Zamalloa et al. 2017), the inaugural sensor-equipped mobile robot, and the widespread adoption of programmable logic controllers (PLCs) honed precision and reliability. The shift from hydraulic to electric drives ensured finer machine control, with microcomputer-guided robots like T3 (Cutkosky and Wright 1982) amplifying operational capacities significantly.

The evolution from the late twentieth century onwards, amplified by AI integration, ushered in a new age for robotics. Machines evolved into intelligent entities capable of learning, autonomous decision-making, and environmental adaptability (He et al. 2017). In the era of Industry 4.0, the fusion of advanced AI algorithms with robotics promises sector-wide transformation, elevating efficiency, curbing expenses, and instigating a fresh wave of innovation.

Contemporary manufacturing deploys a varied robot lineup tailored to distinct production facets. Articulated robots mimic human arm-like flexibility for tasks demanding broad motion spectra with precision. SCARA robots are valued for their speed and accurate performances, making them ideal for assembly lines. Delta robots specialize in swift pick-and-place tasks, ideal for high-speed operations in sectors like electronics and food processing. Cartesian robots, with precise linear motion along three axes, are widely used in applications such as 3D printing and CNC machining (Singh and Banga 2022).

Besides classic robotic arms, the manufacturing domain now witnesses the advent of innovative robotic technology. Automated guided vehicles (AGVs), collaborative robots (cobots), and chatbots spearhead advanced intelligent automation, underlining the Industry 4.0 renaissance. These avant-garde solutions, promising further manufacturing metamorphosis, will be elaborated upon in subsequent sections, underscoring their transformative potential.

4.3.2 Cobots in Manufacturing

Under the Industry 4.0 paradigm, robots are expected to become more flexible, safer, and cost-effective, while also becoming more space-efficient. This evolution is expected to facilitate the integration of robots into work environments, creating new opportunities for human–robot collaborations (Weckenborg and Spengler 2019). These expectations have led to the emergence of cobots, which assist workers in their tasks and improve outcomes and working conditions by leveraging the combination of workers' skills and robots' physical strength and endurance.

Cobots are robots designed with the fundamental concept of human–robot interaction (Colgate et al. 1996). A distinguishing feature of cobots is their incorporation of safety mechanisms, allowing them to operate in close proximity to human operators without posing significant risks. Cobots are equipped with force-limited joints to prevent excessive force application in case of accidental contact with a human worker. Moreover, they are also programmed to operate at reduced speeds when working near humans, minimizing the likelihood of injuries.

The primary advantage of cobots lies in their adaptability to a wide range of manufacturing tasks. These robots can be configured in humanoid or integrated articulated forms and are equipped with force sensors, vision systems, grippers, and other components for control and perception (Cherubini et al. 2016). Their modular design enables swift reconfiguration for various tasks or production lines, such as assembly, pick-and-place, and machine tending, enhancing overall flexibility and efficiency. The drag-and-drop teaching method for programming cobots has stream-

lined operator efficiency and production line flexibility, simplifying the programming process and making it more user-friendly.

Noteworthy distinctions between cobots and traditional robots include the following: (i) cobots operate at lower speeds to ensure a safer working environment for human interaction; (ii) cobots are designed to collaborate with human workers rather than replace them; and (iii) cobots offer lower production and deployment costs compared to traditional industrial robots, along with increased flexibility. In addition to these advantages, cobots also provide ergonomic benefits for repetitive or hazardous tasks and offer opportunities to assist workers with physical disabilities or aging workforces (Keshvarparast et al. 2023).

4.3.3 AGVs in Manufacturing

Another increasingly popular type of robotic systems in manufacturing is the AGV, which autonomously transports materials within a production facility without human intervention. In manufacturing settings, a substantial volume of raw materials or finished products needs to be moved to various locations like warehouses, distribution centers, or manufacturing facilities. These driverless vehicles are adept at receiving transportation tasks, autonomously planning routes, navigating, and executing transportation tasks, leading to significant labor savings and enhanced production efficiency.

The typical operation of an AGV comprises three key components: navigation, steering control, and scheduling (Vis 2006). Navigation is central to AGV systems, encompassing functions such as localization, path planning, and obstacle avoidance. It enables AGVs to accurately determine their own positions and plan the optimal paths to reach the target locations based on task requirements (Huang et al. 2023). Navigation systems range from simple magnetic tape systems to sophisticated sensor-based navigation systems using AI for environmental navigation. To ensure precise navigation and reaching the destination, stable and precise control is imperative. Steering control systems include differential steering control, steering wheel control, and a blend of both, with the integration of control algorithms like proportional-integral-derivative (PID) control, fuzzy

control, and neural network control significantly enhancing control stability (Ryck et al. 2020). Scheduling is critical for assigning specific transport tasks to designated AGVs in complex environments, taking into account constraints such as path conflicts to optimize production efficiency.

An AGV's typical operational cycle involves following steps: 1) when a material transport task arises, the scheduling system assigns the job to the most suitable AGV; 2) Upon receiving instructions, the AGV's navigation system calculates a route leveraging the preloaded facility map and its current coordinates; 3) The AGV's control system propels and steers it to the designated destination; 4) Upon arrival at the designated load station, the AGV precisely halts, triggering the loading mechanism to engage and transfer the cargo; 5) Post loading, the AGV sets off toward the designated offload zone; 6) Completion of the unloading process marks the end of the cycle, at which point the AGV parks itself at a specified location or retreats to a standby zone, primed for the subsequent task.

With the transformation of Industry 4.0, the application of AGV is continuously expanding and becoming more intelligent by integrating advanced AI technologies (Cupek et al. 2020). For example, combining collaborative robots (Cherubini et al. 2016) with AGV unites the dynamic logistic benefits of AGV with the cobots' flexibility, allowing for more extensive use between many assembly stations. AI-driven analytics provides AGVs with powerful analysis and decision-making capabilities, such as path planning and AGV scheduling. In general, the development trends of AGVs are toward intelligence, collaboration, adaptability, versatility, data-driven capabilities, and flexibility.

4.3.4 Chatbots in Design/Manufacturing

A chatbot specifically created to address the requirements and difficulties of the manufacturing sector is known as a manufacturing chatbot. It is a computer program that can converse with users via text or voice, usually in the context of manufacturing and is supported by generative AI and natural language processing technology (Wang et al. 2023). Chatbots designed for the manufacturing industry are trained to understand and execute specific requests, instructions, and tasks related to the design

process, production process, equipment maintenance, quality control, supply chain management, and other manufacturing-related tasks. These AI-driven communication tools can facilitate the flow of information between different entities in the manufacturing process, from the shop floor to the top management (Chen et al. 2021).

One major application is product design a process that typically involves multiple iterations. This process can be time-consuming and resource-intensive. Chatbots can provide real-time feedback and suggestions based on predefined design principles and guidelines, making it faster and easier for designers to reference design databases and retrieving specifications, or for support with design queries, effectively decreasing lead times and enhancing productivity (Dale 2016). Chatbots also facilitate automated customer support, enabling manufacturers to handle routine support queries, service tickets, and complaints process smoother and more proactive (George and George 2023). Additionally, chatbots can provide personalized product recommendations to different customers, enabling manufacturing companies to use their chatbots to assist in shortlisting and making purchasing decisions. Leveraging intelligent algorithms, chatbots can rely on historical data and ML to understand the customer's role and recommend suitable products based on their preferences. Furthermore, predictive maintenance, quality control, and real-time data analysis are areas where chatbots excel, minimizing downtime, enhancing product quality, and optimizing resource utilization. Moreover, chatbots aid in streamlining supply chain management by offering real-time insights, automating inventory control, and improving demand forecasting (Rane 2023).

Currently, manufacturing chatbots is still in an exploratory phase and face several challenges. One notable issue is the lack of transparency in their decision-making processes, leading to occasional inaccuracies in their explanations. Moreover, these chatbots may exhibit constraints in creativity and occasionally present fabricated information in their responses (Wang et al. 2023). Nevertheless, as AI technology progresses, chatbots are expected to become more conversational and intelligent in the future. They are likely to establish deeper integrations with manufacturing systems, including Customer Relationship Management (CRM),

Enterprise Resource Planning (ERP), and Manufacturing Execution System (MES). By incorporating augmented reality, virtual reality, and other complementary capabilities, chatbots will become more sophisticated, adding value to the design and manufacturing sector.

4.4 AI in Manufacturing

AI has emerged as a key driver of transformation in the manufacturing industry, redefining the boundaries of efficiency, quality, and flexibility. This section highlights AI's pivotal role in manufacturing, including how AI empowers predictive maintenance to minimize downtime, enhances quality control through sophisticated analytics, and optimizes production scheduling for maximum throughput.

4.4.1 Integration of AI with Manufacturing Systems

Industry 4.0 and smart manufacturing represent a paradigm shift in the manufacturing landscape, driven by the integration of cutting-edge technologies (Solanki 2023). The most crucial integration among them is the integration of AI technology with manufacturing systems. Encompassing ML, neural networks, deep learning, and other self-organizing systems, AI is capable of rapidly discovering significant patterns in volumes of data that would be beyond the capacity of human analysts (LeCun et al. 2015). This ability to rapidly analyze and learn from large volumes of data is highly applicable to industrial systems, where there is a wealth of historical industrial information and real-time sensor data to analyze and process, such as environmental data, process data, production operation records, and product quality data (Arinez et al. 2020). Therefore, AI has been widely used in the manufacturing industry since its inception and continues to facilitate the development of new models, methods, system architectures, and technological systems in the domain of smart manufacturing, which emphasizes the use of intelligent technologies to optimize production processes and achieve higher levels of efficiency, productivity, and flexibility (Yao et al. 2019a).

The integration of AI with manufacturing systems has evolved from intelligent manufacturing, driven by the combination of knowledge-driven AI

1.0 and traditional manufacturing, to smart manufacturing, characterized by the integration of data-driven AI 2.0 and advanced manufacturing modes like CPS (Yao et al. 2017). This evolutionary trend demonstrates that the integration of AI and manufacturing systems is influenced by both AI technology and the manufacturing systems themselves. From the perspective of manufacturing systems, the shift from traditional industrial models to the emerging manufacturing paradigm of Industry 4.0 has resulted in a more advanced infrastructure. The incorporation of new technologies such as the IoT, CPS, cloud computing, big data analytics, and information and communications technology (ICT), coupled with the widespread utilization of sensors and intelligent robotics, has created numerous entry points for the integration of AI. From an AI perspective, AI 2.0, with advanced technologies such as big data-based intelligence, internet crowd intelligence, cross-media intelligence, and human-machine hybrid-augmented intelligence, is capable of solving problems in current manufacturing systems that are difficult for humans to handle, leading to improvements in efficiency, flexibility, and responsiveness (Pan 2016). Meanwhile, the multitude of demands arising from growing product and process complexity, higher variability in customer demand and preferences, and the increasingly dynamic and complex manufacturing environment have compelled current manufacturing systems to adopt AI to tackle these challenges (Arinez et al. 2020).

In essence, the amalgamation of AI with manufacturing systems forms a symbiotic progression that harmonizes with the evolving patterns of the manufacturing sector. AI has evolved into an essential element of manufacturing systems within the realm of Industry 4.0. The convergence of AI with manufacturing systems entails a methodical framework consisting of three stratified domains: social computing within the social system, cloud computing within the cyber system, and edge computing within the physical system, as depicted in Figure 4.2. Central to this revolution is the collaboration between cloud computing and edge computing, enabling real-time data processing and analysis across various nodes in the manufacturing ecosystem. The harmonious interplay of cloud computing, edge computing, and social computing not only enhances manufacturing operations but also cultivates agility, adaptability, and ongoing enhancement within industrial contexts. This integrated strategy enables businesses to

leverage the complete capabilities of interconnected technologies and human perspectives, fueling innovation, productivity, and competitiveness in the dynamic realm of smart manufacturing.

4.4.2 ML Applications in Predictive Maintenance

The history of manufacturing is intricately linked to the continuous improvement of machine equipment reliability to minimize unexpected downtime, a factor closely tied to maintenance (Endrenyi et al. 2001). Maintenance is a crucial aspect of operating all manufacturing processes, accounting for 15–60% of the total operating costs (Haarman et al. 2017). It has attained critical importance for industries, due to the growth in complexity of the interactions between different production activities in increasingly extended manufacturing ecosystems (Sezer et al. 2018). Predictive maintenance (PdM) is a recent preventive maintenance approach that includes regular monitoring of the mechanical condition, operating efficiency, and other indicators of machine-trains and process systems to gather data for predicting trends, behavior patterns, and correlations, enabling the anticipation of potential failures in advance and enhancing maintenance decision-making to prevent downtime (Haarman et al. 2017; Mobley 2002). It has seen a sharp increase in reliability and effectiveness, becoming a popular maintenance method in smart manufacturing, due to the data-driven ML methods facilitated by the integration of AI and manufacturing systems (Achouch et al. 2022).

PdM aims at establishing a predictive model based on the collection of a large amount of sensor data to predict the temporal progression of machine performance degradation, from its current state to final functional failure. This predictive model is established through the utilization of three primary methods: statistical approaches, AI techniques, and model-based methodologies (Jardine et al. 2006). With its ability to handle high-dimensional and multivariate data, as well as extract hidden relationships within data in complex and dynamic environments, AI techniques, mainly ML, have become the most effective and popular predictive modeling approach (Wuest et al. 2016).

ML algorithms are categorized into three different types: (i) supervised learning, (ii) unsupervised learning, and (iii) reinforcement learning (RL). Among them, supervised learning (e.g., classification, regression) and unsupervised learning (e.g., clustering) are the most commonly used ML methods in PdM. ML algorithm development covers historical data selection, preprocessing data, model selection, model training, model validation, and maintenance (Çınar et al. 2020). Supervised learning techniques such as random forest (RF), artificial neural network (ANN), support vector machine (SVM), and logistic regression often require the collection of a large amount of data including measurements such as vibration, force, torque, accelerometer data, and pressure for both failure status and health condition scenarios in order to effectively train the models. RF is a supervised learning algorithm utilized for classification and regression tasks, known for its strong performance particularly in scenarios where the number of variables exceeds the number of samples. It is the most widely used and compared ML methods in PdM applications for dynamically generating, running process monitoring, and predictive fault detection (Canizo et al. 2017). ANN is another powerful predictive tool, especially with the development of various deep models such as convolutional neural network, recurrent neural network, and long short-term memory (Carvalho et al. 2019). These architectures possess strong nonlinear modeling and time series modeling capabilities, and are widely applied in condition-based maintenance, prognostics and health management, and remaining useful life prediction (Liu et al. 2020). Unsupervised learning algorithms like K-means primarily focus on identifying clusters within data without the need for feedback from an external teacher or knowledgeable expert. It is commonly used to identify clusters in the data and classify types of faults (Durbhaka and Selvaraj 2016).

It has been observed that ML techniques are increasingly being applied in designing PdM applications, leading to minimized downtime, increased utilization rate of components, and extended remaining useful lives. However, there are still some aspects in PdM and ML that require further investigation, such as choosing the most appropriate ML algorithms for a proposed PdM strategy, proposing multiple ML methods (ensemble learning) in a PdM application, and creating new datasets to be used and compared in PdM research.

4.4.3 ML Models in Quality Control

Quality control is an important part of the production process that ensures products meet expected standards and minimizes the occurrence of defects and nonconforming items (Mitra 2016). It has become increasingly important due to increased customer demand for higher quality products, an increasing emphasis on sustainable production, and rapidly rising market competition in smart manufacturing (Colledani et al. 2014). Therefore, a new production quality control system is needed for mass customization (Daaboul et al. 2011), mass personalization, and high product variety manufacturing (Yao et al. 2017). ML, within AI, is a new and effective way to address quality control challenges, leading to improved product quality, continuous evaluation, and adaptation to meet evolving customer needs, ultimately enhancing productivity (Escobar and Morales-Menendez 2018). By utilizing ML algorithms to analyze various sensor data such as dimensional measurements, speed, vibration, and visual data, the quality control system can effectively minimize defect production and efficiently achieve real-time defect identification.

Monitoring and controlling manufacturing processes are fundamental in quality assurance, ensuring consistency and stability by adjusting parameters swiftly to maintain quality. Real-time sensor data analysis is crucial in process control for adapting promptly to the dynamic nature of manufacturing. Traditional statistical methods like SPC address deviations but struggle with dynamic changes (Singer and Cohen 2021). ML excels by modeling nonlinearities, using historical data to predict process variations early and forecast errors, reducing faults and enabling agile responses. ML applications span fast-paced manufacturing like laser cutting, metalwork, grinding, and welding. For instance, ANNs trained with key parameters predict laser cutting quality through nonlinear modeling (Chaudhari et al. 2012). ML algorithms such as RF, SVMs, and ANNs enhance grinding control by analyzing features from the grinding wheel's condition (Lv et al. 2020). By integrating ML, industries can optimize monitoring and control effectively across various manufacturing processes.

The process of defect detection involves inspecting and testing products at every stage of production to identify and mitigate potential defects,

with the goal of achieving quality control objectives. Traditional approaches to defect detection often rely on manual visual inspections or specialized testing equipment, which are susceptible to human errors and are inefficient in nature. However, by leveraging automated sensor measurements and ML algorithms for autonomous decision-making, the speed and accuracy of defect detection can be significantly enhanced. From a ML standpoint, the task of defect detection can be formulated as a multi-class classification problem, which can be effectively tackled through the utilization of decision trees, RF, SVM, and other efficient ML algorithms, in conjunction with effective feature engineering techniques (Escobar and Morales-Menendez 2018). Machine vision algorithms play a crucial role in defect detection by providing easy deployment, minimal feature engineering requirements, and rapid detection speeds, as most defects are visible to the naked eye. Traditional ML methods primarily rely on collecting texture features, color features, and shape features of workpieces through cameras for classifier training. This approach depends on the design and accurate extraction of features, while also having strict requirements and potential biases in the state of the images. With the rapid development of deep convolutional neural networks (DCNNs), and the introduction of powerful architectures such as ResNet, Faster RCNN, YOLO and fully convolutional networks, the field of visual defect detection has experienced significant breakthroughs based on DCNNs (Krizhevsky et al. 2012). Neural networks pre-trained on defect detection datasets can accurately pinpoint the location and type of defects, showcasing robust generalization abilities. While these networks heavily depend on dataset quality and quantity, necessitating extended training periods, challenges can be effectively addressed through methods like data augmentation and transfer learning. Today, defect detection technologies based on DCNNs are widely used across various materials such as wood, steel, ceramic, patterned fabric, etc., for detecting a diverse range of defects including cracks, scratches, offsets, welding defects, and 3D printing defects (Caggiano et al. 2019).

4.4.4 Optimizing Production Scheduling with AI Algorithms

Production scheduling is a decision-making process in manufacturing that involves effectively allocating resources to activities over time to sat-

isfy temporal and resource capacity constraints, optimize various objectives, and ultimately meet the needs of both customers and manufacturers (Fazel Zarandi et al. 2020). It has evolved from the initial single-machine scheduling problem to complex dynamic scheduling problem in the context of Industry 4.0 smart manufacturing and mass customization requirements (Parente et al. 2020). Traditional exact and heuristic solving methods suffer from slow computation times, poor generalization, and reliance on human expert knowledge, making them ineffective in addressing the increasingly complex scheduling problems arising in smart manufacturing such as dynamic flexible shop scheduling, distributed shop scheduling, and integrated multi-resource shop scheduling (Li et al. 2022). ML methods, especially neural networks and deep reinforcement learning (DRL), can utilize large amounts of data to learn complex manufacturing scheduling knowledge. By integrating constraints, state information, and heuristic methods, they offer rich, structured knowledge representation, enabling fast and accurate optimal decision-making to optimize production processes and improve efficiency (Shaw et al. 1992).

The Hopfield network, consisting of a fully connected network of neurons capable of performing computational tasks, is one of the earliest neural network algorithms applied to workshop scheduling (Akyol and Bayhan 2007). In this network, both the objective function and problem constraints are encoded using a suitable energy function with the aim of achieving a configuration that minimizes the energy function. The job shop scheduling problem can be addressed by utilizing a two-dimensional Hopfield TSP-type matrix of neurons to find the global minimum of the energy function (Simon, 1988). Another important type of network used in production scheduling is the multilayer perceptron, a feedforward network consisting of neurons connected by weighted links. By establishing appropriate weights between the operational policy of a work center and performance measures such as mean costs, mean flow time, and mean tardiness, an effective production schedule can be achieved, extending to dynamic scheduling problems (Chryssolouris et al. 1991).

In recent years, DRL has shown powerful solving capabilities for production scheduling by combining the strong state representation and learning abilities of deep learning with the exploration and decision-making

capabilities of RL (Li et al. 2023). To solve production scheduling problems using DRL, the problem is typically modeled as a Markov decision process, which requires the determination of an environmental state representation, actions, state transitions, and reward functions. The state of workshop scheduling typically encompasses the collective state of resources, including workpieces and machines, represented as matrices incorporating their individual and overall characteristics. Graph neural networks have become a popular method for representing workshop states due to their powerful embedding capabilities and compatibility with traditional disjunctive graph representations (Wu et al. 2020). By propagating information on nodes representing entities such as workpieces and machines, along with edges representing constraints, workshop information can be effectively embedded into respective nodes (Song et al. 2022). Combined with efficient actions selection networks like pointer networks and attention mechanisms, efficient scheduling can be achieved (Park et al. 2021). Actions can involve selecting a particular workpiece to allocate to a machine or choosing a specific scheduling rule to apply. State transition involves determining the environment changes following actions, while the reward function defines the reward value obtained after taking actions. After establishing the Markov model, it is crucial to choose a suitable RL to train algorithm, which includes value-based, policy-based, and actor-critic algorithms. The most commonly used is the Actor-Critic algorithms, with the most mainstream being the proximal policy optimization algorithm, as it combines the advantages of both value-based and policy-based methods, making it more efficient (Schulman et al. 2017). Additionally, it is worth mentioning that modeling the workshop system as a multi-agent system and using multiagent RL algorithms can enable distributed execution to reduce computational load and simplify the design of state and action spaces (Jing et al. 2024). However, deploying and training multi-agent algorithms can be challenging, as they are prone to convergence issues and require further research.

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